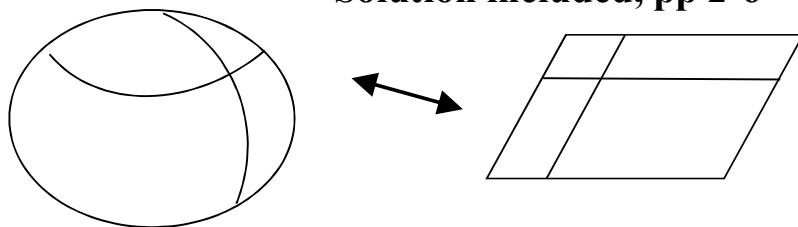


TBA4245 GEODESY

Example exercise no. 4: The Gaussian conformal projection.

Solution included, pp 2-6



(1) The general Gaussian conformal projection.

From known latitude and longitude (B, L) on the spheroid to coordinates in the projection plane (x, y)

Known latitude and longitude (in the EUREF89 datum) of point P

EUREF89-Geographic

Latitude, B : $63^\circ 24' 53.9692'' = 63.41499144^\circ$

Longitude, L : $10^\circ 24' 22.8294'' = 10.40634150^\circ$

Calculate the UTM coordinates from the known geographic coordinates (B and L).

Actual questions:

The formulas given in the text book from **Hofmann-Wellenhof**: “GNSS” has 1 or 2 extra terms compared to the formulas in “Geodesy part 2” (the old version). In which cases do we need the extra terms to keep up the accuracy of the coordinates (result)?

Which adjustments must be done to obtain UTM coordinates from the coordinates of the Gaussian conformal projection?

Which adjustments must be done to obtain NGO1948 coordinates from the coordinates of the Gaussian conformal projection? And NTM coordinates?

(2) The general Gaussian conformal projection.

From known coordinates in the projection plane (x, y) to latitude and longitude (B, L) on the spheroid

Known UTM coordinates (in the EUREF89 datum) of point P

EUREF89-UTM $N = 7032598.406$ $E = 570217.471$ Zone 32

Calculate the geographic coordinates (latitude B and longitude L) from the given UTM coordinates.

Actual question:

The formulas given in “GNSS” has 1 or 2 extra terms compared to the formulas in “Geodesy part 2”. In which cases do we need the extra terms to keep up the accuracy of the coordinates (result)?

Example exercise no. 4. Solution

(1) The general Gaussian conformal projection.

From known latitude and longitude (B, L) on the spheroid to coordinates in the projection plane (x, y)

EUREF89. Which UTM zone? Our point is ca. 10° east of Greenwich. Zone 32 is between 6 and 12 degrees east of Greenwich. The east value of the Central meridian (= the north axis) in the middle of the zone: $L_0 = 9^\circ$ east of Greenwich.

We have to find the longitude l on figure 4.3 in Geodesy part 2. The angle l is the angle from the Central meridian to the meridian through the point ("local" longitude).

$$l = 10.40634150 - 9 = +1.40634150^\circ$$

We need the principal radius of curvature, $N = R_N$, in the formulas:

$$N = \frac{a^2}{b \cdot \sqrt{(1 + \eta^2)}} = 6395278.878$$

$$\eta^2 = e'^2 \cdot \cos^2 B = 1.34977785409 \cdot 10^{-3}$$

$$e'^2 = \text{the 2nd numerical eccentricity in square} = \frac{a^2 - b^2}{b^2}$$

The general Gaussian projection formulas:

$$x = G - a_2 \cdot l^2 + a_4 \cdot l^4 - a_6 \cdot l^6 + \dots \quad y = a_1 \cdot l - a_3 \cdot l^3 + a_5 \cdot l^5 - a_7 \cdot l^7 + \dots$$

Calculating the north coordinate in the map projection plane (Gaussian conformal projection, GCP):

Spheroidal distance from the Equator to the actual point: Here the Holsen program MERIDBUE is used, formulas are found in "Geodesy part 1" and in chapter 8 in "GNSS". $G = 7034641.566$ m

The coefficients:	The terms
G	$G = 7034641.566$ m
$a_2 = -\frac{1}{2} \cdot N \cdot \cos^2 B \cdot \tan B = -1279723.770$	
	$a_2 \cdot l_{\text{radian}}^2 = -770.9968$ m
$a_4 = \frac{1}{24} \cdot N \cdot \cos^4 B \cdot \tan B \cdot (5 - t^2 + 9\eta^2 + 4\eta^4)$ $a_4 = 21358.45433 \cdot 1.01911385037 = 21766.69664$	
	$a_4 \cdot l_{\text{radian}}^4 = 0.0079$ m
$a_6 = -\frac{1}{720} \cdot t \cdot N \cdot \cos^6 B \cdot (61 - 58t^2 + t^4 + 270\eta^2 - 330t^2 \cdot \eta^2)$ $a_6 = -142,588 \cdot (-156,06619) = 22253,1874$	
	$a_6 \cdot l_{\text{radian}}^6 = 0.000005$ m
Adding the 4 terms:	$x_{\text{GCP}} = \mathbf{7035\ 412.5707}$

Calculating the east coordinate in the map projection plane (Gaussian conformal projection, GCP):

The coefficients:	The terms
$a_1 = N \cdot \cos B = 2862047.9733$	
	$a_1 \cdot l_{radian}^1 = 70249.7963 \text{ m}$
$a_3 = -\frac{1}{6} \cdot N \cdot \cos^3 B \cdot (1 - t^2 + \eta^2) = 285809.9 \quad t = \tan B$	
	$a_3 \cdot l_{radian}^3 = 4.2265 \text{ m}$
$a_5 = \frac{1}{120} \cdot N \cdot \cos^5 B \cdot (5 - 18t^2 + t^4 + 14\eta^2 - 58t^2 \cdot \eta^2) = -48855.5$	
	$a_5 \cdot l_{radian}^5 = -0.0004 \text{ m}$
Adding the 3 terms:	$\Rightarrow y_{GCP} = 70245.5694$

Adjusting to UTM coordinates:

$$N = x_{GCP} \cdot 0.9996 = 7035\,412.5707 \cdot 0.9996 = 7032598.4057$$

$$E = y_{GCP} \cdot 0.9996 + 500000 = 70245.5694 \cdot 0.9996 + 500000 = 570217.4711$$

From the official program WSKTRANS:

$$N = 7032598.406 \quad E = 570217.471 \quad (\text{Zone } 32) \Rightarrow \text{OK!!}$$

When do we need extra terms in the formulas in Geodesy part 2?

See Hofmann-Wellenhof: "GNSS" chapter 8.2 for the projection formulas with some extra terms.

When $l = 3^\circ$ (and $B = B_P = 63.41499144^\circ$) the numerical values of the extra terms are:

$$a_6 \cdot l_{radian}^6 = 0.0005 \text{ m} \quad a_5 \cdot l_{radian}^5 = -0.019 \text{ m}$$

Remember the coefficients are functions of the latitude. In the formulas the cosine will decrease and the tangent will increase with increasing latitude (no computations showing the effects on the numerical values of the terms...)

The east value is mostly dependent of l , the difference in longitude from the Central meridian to the meridian through the point $\Rightarrow$ a large numerical value of the longitude l demands more terms to keep up the accuracy. See the example using a point at the Equator shown below.

The north value? Compute the terms when the point has a latitude of $B = 71^\circ$ (ca North cape) and latitude $L = 24^\circ$. For the answers, see in the next version of the solution (!?)

Example, point P on the Equator ($B = 0^\circ$ and $L = 12^\circ$, thus the local latitude is $l = 3^\circ$)

The north coordinate: $x = 0$

The east coordinate: $y = a_1 \cdot l - a_3 \cdot l^3 + a_5 \cdot l^5 - a_7 \cdot l^7$

$$= 333958.4724 + 153.6228 + 0.1066 + 0.00008 = 334112.202$$

The UTM coordinate: $E_{UTM} = 0.9996y + 500000 = 833978.557$

The scale factor, cf. example exercise no 5:

$$m = 1 + 0.0013800168 + 0.000001566 = 1.00138158$$

$$m_{UTM} = 0.9996m = 1.000981003, \text{ which is greater than } 1.0004 \text{ (but is OK)}$$

Calculations NGO1948 coordinates? Similar for NTM coordinates, check the longitude of the north axis.

- The Central meridian of zone 3 is 10.72291667° east of Greenwich, see a formula page in the preface of Geodesy part 2. From this longitude we calculate a corresponding value of the local longitude, l : $l = 10.40634150 - 10.72291667 = -0.31657517^\circ$ (is negative!)
- The origin is at 58° north. Therefore we must subtract the arc length of the meridian from Equator to 58° north: $G = 6430707.920$ m
- No correction of the scale along the Central meridian.

With the known EUREF89-UTM values we obtain the following NGO1948 coordinates by using the official transformation program WSKTRANS:

NGO1948 $N = 603286.853$ $E = -15548.032$ Axis no. 3

NB: But, the numerical values of latitude and longitude in the NGO1948 datum are not identical with the corresponding values of the latitude and longitude in the EUREF89 datum. Why?

**(2) The general Gaussian conformal projection.
From known coordinates in the projection plane (x, y) to latitude and longitude (B, L) on the spheroid**

Known UTM coordinates (in the datum EUREF89) of point P.

EUREF89-UTM $N = 7032598.406$ $E = 570217.471$ Zone 32

Calculate the geographic coordinates (latitude B and longitude L) from the given UTM coordinates.

Answer:

Must have GCP coordinates, not UTM coordinates. Gaussian conformal coordinates in the map projection plane of point P, see question (1) for the method used from GCP to UTM values:

$$\text{North: } x_{GCP} = N / 0.9996 = 7032598.406 / 0.9996 = 7035412.571$$

$$\text{East: } y_{GCP} = (E - 500000) / 0.9996 = (570217.471 - 500000) / 0.9996 = 70245.569$$

The Taylor series is extracted in the foot point P_F at the Central meridian. Point F has the same north coordinate in the plane as point P. **We therefore have to find the latitude of the foot point P_F : B_F .**

Here we use the formula 8.27 (and 8.28-8.30) in “GNSS”.

Transforming the north coordinate, where the value of α is obtained from a table:

$$\bar{y}_{GPSBOOK} = \frac{y_{GPSBOOK}}{\alpha} = \frac{7035412.571}{6367449.1458} = 1.10490282842$$

We need the principal radius of curvature, N , in the foot point P_F for use in the formulas:

$$N_F = \frac{a^2}{b \cdot \sqrt{1 + \eta_F^2}} = 6395278.878$$

$$\eta_F^2 = e'^2 \cdot \cos^2 B_F = 1.34912669366 \cdot 10^{-3}$$

$$e'^2 = \text{the 2nd numerical eccentricity squared} = \frac{a^2 - b^2}{b^2}$$

$$\tan B_F = t_F = 1.99886251291$$

The formula (8.27) gives the latitude B_F term by term:

$$\begin{aligned}
 B_F = & 1.10490282842 \\
 & +0.00202182932 \\
 & -0.00000354346 \\
 & +0.00000000253 \\
 & +0.00000000001 = 1.10692111682 \text{ radians} = 63.421908248^\circ
 \end{aligned}$$

Checking by using the program MERIDBUE, entering the values of point P_F : The north coordinate in the plane = the arc length of the Central meridian (G) (as the scale is 1:1). The results are the latitude of P_F and the arc length of the meridian, which in this case is equal to G :

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*****
**      BEREGNING AV MERIDIANBUEN      **
**                                     **
**      J. H. APRIL 1990                **
*****

R SETTES LIK 0 NÅR <B1,B2> ER GITT .I MOTSATT FALL
SETTES R LIK 1 NÅR B1 OG G ER GITT
1
VALG AV ELLIPSOIDE.P SETTES LIK 1 FOR:
NORSK BESSELS, LIK 2 FOR INTERNASJONAL:
OG LIK 3 FOR WGS-84-ELLIPSOIDEN:
LEGG INN P:
3
LEGG INN B1 OG G:
0
7035412.571
BREDDE B2=63.421908250
MERIDIANBUE = 7035412.571
Stop - Program terminated.

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Calculation of the latitude of point P by using the formula no. 8.26 in the “GNSS” book). This formula contains some extra terms compared to formula (4.50) in the compendium Geodesy part 2:

$$B_P = B_F + b_2 \cdot y_{GCP}^2 + b_4 \cdot y_{GCP}^4 + b_6 \cdot y_{GCP}^6 + b_8 \cdot y_{GCP}^8 + \dots$$

Term	Value in radians	Value in degrees
B_F	1.10692111682	+63.421908248°
$b_2 \cdot y_{GCP}^2$	$-2.446916672 \cdot 10^{-14} \cdot 70245.569^2 = -1.20741634 \cdot 10^{-4}$	-0.006917986°
$b_4 \cdot y_{GCP}^4$	$+8.4452228 \cdot 10^{-28} \cdot 70245.569^4 = +2.0563018 \cdot 10^{-8}$	+0.000001178°
$b_6 \cdot y_{GCP}^6$	$-4.6148 \cdot 10^{-41} \cdot 70245.569^6 = -5.544 \cdot 10^{-12}$	-0.0000000003°
$b_8 \cdot y_{GCP}^8$	Must become zero here, see the value of the previous term	0
	$\Rightarrow B_P$	= 63.4149914399° = 63.41499144°

Calculation of the latitude of point P, by using the formula 8.26 in the “GNSS” book. This formula contains some extra terms compared to formula (4.50) in the compendium Geodesy part 2:

$$L_P = L_0 + b_1 \cdot y_{EAST}^1 + b_3 \cdot y_{EAST}^3 + b_5 \cdot y_{EAST}^5 + b_7 \cdot y_{EAST}^7 + \dots$$

Term	Value in radians	Value in degrees
L_0		+9.000000000°
$b_1 \cdot y_{EAST}^1$	$+3,4948434881 \cdot 10^{-7} \cdot 70245,569^1 = +2,4549726939 \cdot 10^{-2}$	+1.406595742°
$b_3 \cdot y_{EAST}^3$	$-1,280635894 \cdot 10^{-20} \cdot 70245,569^3 = -4,4389727 \cdot 10^{-6}$	-0.000025433°
$b_5 \cdot y_{EAST}^5$	$+8,706072 \cdot 10^{-34} \cdot 70245,569^5 = +1,489076 \cdot 10^{-9}$	-0.000000085°
$b_7 \cdot y_{EAST}^7$	$-7,064 \cdot 10^{-47} \cdot 70245,569^7 = -5,96226 \cdot 10^{-13}$	-0.0000000003°
	$\Rightarrow L_P$	= 10.40634149°

Checking by using the official program SKTRANS, gives identical results!:

EUREF89-Geo

Latitude: 63° 24' 53.9692" = **63.41499144°**

Longitude: 10° 24' 22.8294" = **10.40634150°**

When do we need extra terms in the formulas in Geodesy part 2?

See Hofmann-Wellenhof: “GNSS”, chapter 8.2 for the projection formulas with some extra terms.

If the point P' is situated at the borderline between zone 32 and 33, and has the same north value, N , as point P ($N = 7032598.406$) and a east value E of 649883.746 (which corresponds to a local longitude of $l = 3^\circ$ and a east coordinate in the Gaussian projection plane equal to $y_{GCP} = 149943,723$ m)

And then the smallest terms get the following numerical values:

$$b_6 \cdot y_{GKA}^6 = -0.0000003^\circ = -0.00011'' \text{ (seconds)}$$

$$b_7 \cdot y_{GKA}^7 = -0.0000007^\circ = -0.00002''$$

Remember also that the values of the coefficients are dependent on the value of the latitude. In the formulas the cosine will decrease in value and the tangent will increase in value as the latitude is increasing.

The value of the longitude is mostly dependent of the value of the east coordinate in the projection plane, y . $\Rightarrow$ a large numerical value of the east coordinate in the plane demands more terms to keep up the accuracy. See the example using a point at the Equator shown below.

The value of the latitude, B : Compute the terms when the point has a latitude of $B = 71^\circ$ (ca North cape) and latitude $L = 24^\circ$. For the answers, see in the next version of the solution (!?). The point has the UTM- coordinates $N = 7880094.920$ and $E = 608970.191$

Example, point P on the Equator ($N = 0$ m and $E = 833978.557$ m , gives $y_{UTM} = 334112,202$ and the latitude $B = 0^\circ$ and the longitude $L = 12^\circ$ (lying in the borderline between the UTM zones 32 and 33), which gives a local longitude of $l = 3^\circ$).

$$\begin{aligned}
 L_P &= L_0 + b_1 \cdot y_{EAST}^1 + b_3 \cdot y_{EAST}^3 + b_5 \cdot y_{EAST}^5 + b_7 \cdot y_{EAST}^7 + \dots \\
 &= 9,000000000 + 3,001380977 - 0,00131923 + 0,000000949 - 0,0000000008 \\
 &= 12,000000001^\circ \text{ OK!}
 \end{aligned}$$